Large-Vocabulary Chord Recognition Based on Contrastive Learning and Noisy Student (2023)

Chen Li, Jingyi Jiang, Yu Li, and Lihua Tian

***Abstract*—Automatic chord recognition can improve the performance of music information retrieval, thereby optimizing the user experience of music electronic products. In order to improve the accuracy of automatic chord recognition, we carry out our research from two perspectives: model structure and training strategies. Firstly, we propose a convolutional neural network with self-attention mechanism as the backbone network. This network well combines the short-time feature capture ability of convolutional neural network with the long-term feature capture ability of self-attention mechanism, to exact better chord features. Furthermore, a semi-supervised automatic chord recognition algorithm is introduced based on contrastive learning and a noisy-student network to address the challenge of insufficient high-quality labeled data. The algorithm comprises two stages, namely representation learning and semi-supervised learning. In representation learning stage, we designed a more refined method to construct for positive and negative sample pairs so as to better learn the characteristics of the entire data set. In the context of representation learning, the semi supervised learning stage utilizes both labeled data and high-quality unlabeled data to enhance the training of the model to improve its performance. The experimental results show that the proposed algorithm significantly improves the accuracy of chord recognition.**

***Index Terms*—Chord recognition, contrastive learning, music electronic products, noisy-student.**

I. Introduction

With music becoming an integral part of daily life, music applications occupy a considerable proportion in the current Internet software market. Through identifying various chords, music similarity calculation can be better completed, and themes and genres of music can be classified to better recommend music and albums that users are interested in [1]. It can be seen that automatic chord recognition (ACR) has optimized the user experience in information entertainment.

Currently, publicly available labeled datasets are very rare in the field of ACR, as it requires professionals with music skills and time-consuming [2], [3]. Hence, it is difficult to improve the accuracy in the supervised domain. In machine learning, semi-supervised learning is a solution to the problem of label sparsity, which can utilize a small portion of labeled data and a large amount of unlabeled data to improve the model generalization performance [4]. Therefore, it can be seen that semi-supervised learning is more promising than supervised learning in the field of chord recognition with limited labeled data.

To address these problems, we propose a semi-supervised automatic chord recognition algorithm from the perspectives of model structure and training strategy, which consists of two phases: representation learning and semi-supervised learning. In representation learning, pairs of positive samples are constructed by data augmentation, and pairs of positive and negative samples are constructed by means of calculating the overlap of the chord labels in the different audio clips. The model learns the general features of the datasets by learning the relation and difference of feature representations between positive and negative samples. In semi-supervised learning stage, the model is trained based on the noisy-student framework by using pseudo-labeled data with higher quality and labeled data. Two phases of training enable the model to fully and comprehensively learn from unlabeled data. Experimental results show that the proposed semi-supervised automatic chord recognition algorithm based on contrastive learning and noisy-student performs well on almost all evaluation metrics.

The key contributions of this study can be summarized as follows:

- As for training strategies, a semi-supervised automatic chord recognition algorithm based on contrastive learning and noisy student framework is proposed. For the first time, contrastive learning is introduced into the field of automatic chord recognition. Through the representation learning stage and semi-supervised learning stage, unlabeled data is comprehensively and fully utilized, maximizing the advantages of a large amount of unlabeled data.
- Based on the unique data augmentation method, a method of constructing positive and negative sample pairs by using pseudo-label and calculating the overlap with the chord labels in the different audio clips is proposed, so that the model can learn more features.
- From the perspective of model structure, a three-stage model is proposed, which can combine convolutional neural networks which has strong ability to capture short-term features with self-attention mechanism which has strong ability to capture long-term features. And in the following decoding, conditional random fields (CRF) is adopted to capture more features and obtaining smoother results that is more in line with music theory.

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The rest of this paper is as follows: the related work about automatic chord recognition and the theory of the purposed algorithm in this paper are introduced in Section II. In Section III, a detailed description of the model and the algorithm proposed in this paper is provided. As for Section IV, the experimental results and analysis are shown in this section. While in Section V, we draw the conclusion and clarify the future work.

II. Related Work

A. Automatic Chord Recognition

The earliest research on automatic chord recognition can be traced back to the 1990s when Fujishima [5] proposed a real-time automatic chord recognition system and quickly made it a major research topic in the International Conference on Music Information Retrieval (ISMIR) community. Until now, automatic chord recognition continues to support consumer-level tasks such as music classification [6] and music recommendation [7]. Early algorithms for automatic chord recognition relied on pattern matching, which compared the twelve-dimensional chromatic features of the input music with a binary system of chord dictionary, and identified the most similar chord template as the output. With the rise and development of machine learning and deep learning, automatic chord recognition algorithms have gradually transitioned from being knowledge-driven to data-driven.

The main issue faced by the field of automatic chord recognition currently is the problem of imbalanced classification [8]. This leads that the recognition accuracy of rare chords is much lower than common chords. It is mainly due to the scarcity of publicly available labeled datasets and the pitiful small sample size of rare chords in the current field of chord recognition, since the datasets of ACR is time-consuming and requires expertise in music. Such a small number of samples makes the training ineffective.

To address the problem of the scarcity of the labeled data, research in the field of chord recognition has mainly focused on two directions. The first is based on the training strategy. Luke Rowe and George Tzanetakis apply curriculum learning to ACR tasks [9], continuously increasing the weight of rare chords during the training process and facilitating the model to pay more attention to rare chords in the middle and later stages of training. The second is to introduce more unlabeled datasets to expand the datasets. Marcelo Bortolozzo et al. used a self-learning approach to generate pseudo-labels for a large amount of unlabeled data using a pre-trained model, and reconstructed a more balanced datasets based on the pseudo-labels for model training [10]. References [9], [10] these methods significantly improve the recognition accuracy of rare chords, while losing the accuracy of common chords, thereby causing significant damage to the overall recognition accuracy. As for in semi-supervised area, Wu et al. proposed a semi-supervised ACR algorithm based on the variational autoencoder [11]. This method is a good combination of discriminative and generative methods. They applied features extracted by a convolutional neural network that is trained with Music Instrument Digital Interface (MIDI) data as intermediate variables. All subsequent training relies on the pitch level activation features extracted by the convolutional feature extractor [12], which generalization is insufficient. Based on [11], Wu et al. proposed a multi-task hierarchical variational autoencoder algorithm that integrates chord and key information. The model is trained in a semi-supervised manner and used for music generation [13], which improves automatic chord estimation but suffers from the same drawbacks as [11]. Therefore, there is an urgent need in ACR for a simple and general semi-supervised approach to enhance the accuracy of chord recognition.

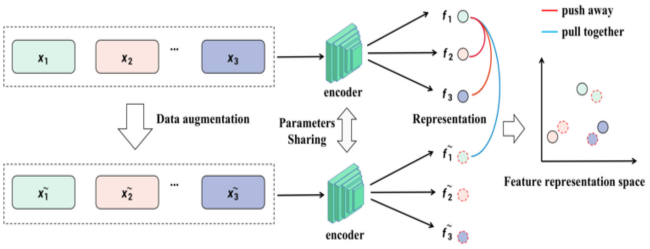


Fig. 1. Overall framework diagram of contrastive learning.

B. Contrastive Learning

Contrastive learning is a self-supervised learning method [14] that does not rely on high-quality labeled data. It has the capability to acquire general features from a dataset in the absence of labels by enabling the model to discern similarities or differences among individual data points within the dataset. The overall framework of contrastive learning is illustrated in Fig. 1.

Contrastive learning is based on a large amount of unlabeled data, making it difficult to ascertain the similarity between any two samples. Therefore, data augmentation is usually used to construct a similar sample to the current one, and then align the feature representations of the original sample with those of the augmented sample. The original sample and the augmented sample constitute a positive sample pair, while the original sample and the remaining samples in the same training batch are negative sample pairs, which is also the most fundamental and simplest way to construct positive and negative sample pairs for contrastive learning. After the positive and negative sample pairs are constructed, it is imperative to impose constraints on the model so as to maximize the difference between the feature representations of negative sample pairs. This approach facilitates the model to learn the general feature representations of the dataset. This is usually realized by using a contrastive loss function (InfoNCE loss [19]) which is defined as a loss function that is used to learn the general features by minimizing feature representations of similar samples and maximizing that of less similar ones, as defined in (1):

$$Loss(x_i) = -log\left[\frac{\exp\left(\frac{sim(x_i,x_i')}{\tau}\right)}{\sum_{k\neq i}^{N}\exp\left(\frac{sim(x_i,x_k)}{\tau}\right)}\right] \tag{1}$$

where x_i denotes the current sample. x_i' denotes the positive sample after random augmentation. $sim(x_i, x_i')$ denotes

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the similarity between the positive sample pairs. τ denotes the temperature coefficient, and the numerator is used to quantify the similarity between the positive samples. N represents the total number of samples in the training batch x_i, and the denominator is used to quantify the similarity between negative samples. It can be seen that the most significant effect of the contrastive loss function is to make positive samples as similar as possible while negative samples as different as possible. The reason why the contrastive loss function performs well is that it is a loss function of self-mining for hard samples. Contrastive loss poses a larger gradient to negative samples that are more similar to the current sample, thus allowing the model to better differentiate these confusing samples. The temperature coefficient determines the extent to which the contrastive loss prioritizes hard samples. The larger it is, the less the contrastive loss function will focus on hard samples. The smaller it is, the more it will focus on negative samples that are highly similar to the current sample. A large temperature coefficient will severely degrade the self-mining function of hard samples in the contrastive loss. While a small temperature coefficient will make the model place more emphasis on highly similar negative samples. The class of these negative samples cannot be accurately known, thus some of the highly similar negative samples are likely to be in the same category as the current sample. However, pushing these samples away from the current sample will severely damage what this model already learned. Therefore, the temperature coefficients are often selected through several experiments, and analyzed based on specific context so as to reach a relatively balanced state.

C. Noisy-Student Network

The noisy-student network is a classical semi-supervised algorithm that has delivered excellent results in melody extraction, song labeling and other music information retrieval tasks in recent years [15], which can significantly improve the recognition accuracy of the model. Firstly, a well-performing teacher model is trained with the labeled data, and then the teacher model is used to generate pseudo-labels for a large amount of unlabeled data. Pseudo labels can be hard or soft labels, and research shows that using soft labels, a form of pseudo labels, will give better results. Subsequently, the unlabeled data are input into the student model together with the labeled data after augmentation. The student model is trained based on the pseudo labels generated by the teacher model and the real annotation. The specific training loss function is shown in (2):

$$Loss_{sl} = \frac{1}{N}\sum_{i=1}^{N} H(y_i, p(y|x_i;\theta_s)) + \frac{1}{M}\sum_{j=1}^{M} H\Big(p(y|x_j;\theta_t), p(y|x'_j;\theta_s)\Big) \quad (2)$$

where N denotes the number of labeled data. H denotes the cross-entropy loss function. y_i represents real annotation corresponding to the labeled sample. θ_s denotes the parameter set of the student model, and $p(y|x_i;\theta_s)$ denotes the prediction of the student model. The first segment of (2) is used to calculate the supervised loss. M denotes the number of high-quality unlabeled data after selection, and $p(y|x_j;\theta_t)$ indicates the prediction of the teacher model towards the unlabeled sample x_j. x'_j denotes the sample after data augmentation for x_j. $p(y|x'_j;\theta_s)$ denotes the prediction of the student model towards the sample x'_j. The samples input to the student model are those with slight perturbations, while what the teacher model inputs are clean. Therefore, the performance of the student model can be improved by fitting the predictions of the student model and teacher model [16].

III. Purposed Method

A. Bidirectional Attention-Augmented Convolutional Neural Network

In terms of model structure, a convolutional neural network is combined with a bidirectional self-attention mechanism in this paper, which performs well in ACR. In music theory chord progressions are highly repetitive. When identifying the chord category at a given moment, it is necessary to focus not only on the feature representation at this moment but also at other moments within this chord progression. Convolutional neural networks focus on feature representation at the current moment, whereas self-attention mechanisms concentrate more on long-time feature representation. Therefore, we augment the convolutional neural network with the bidirectional self-attention, aiming to leverage the capability to capture short-time features of the convolutional neural network and that of capturing long-time features of the bidirectional self-attention. The overall structure of the model proposed in this paper is shown in Fig. 2. The model takes the Constant-Q transform (CQT) spectrogram as input, and contains normal convolutional neural network layers and bidirectional attention-augmented convolutional network layers. Each convolutional layer will perform zero-padding so that the output is consistent with the initial CQT features in the spatial dimension. Each convolutional layer is followed by batch normalization. Then, the output after Rectified Linear Unit (ReLU) is used as the input to the next convolutional layer. Skip connections are conducted between the layers of the common convolutional network. Skip connection is a network design technique introduced in Resnet [17], which makes network training easier by allowing nonlinear transformations to learn the residual mapping instead of direct mapping between input and output. Among the layers of the convolutional neural network augmented by self-attention, the output from the preceding layer is incorporated into that of the normal convolutional neural network and the bidirectional self-attention proposed by [18] respectively, and finally combine their outputs.

B. Overview About the Proposed Algorithm

The key to semi-supervised learning is how to effectively use a large amount of unlabeled data, which is usually started by using the labeled data in the pre-training phase to train a well-performing teacher model. Then the teacher model is employed to generate pseudo-labels for those unlabeled

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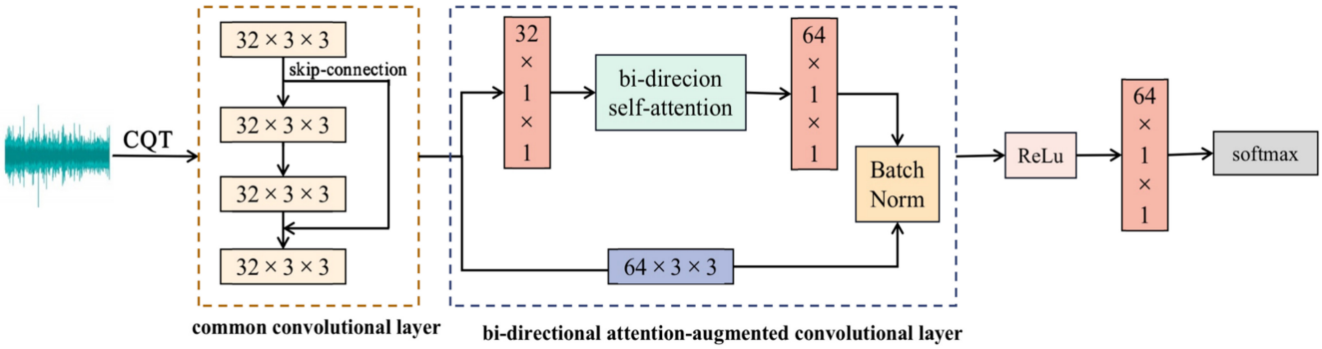


Fig. 2. Bidirectional attention-augmented convolutional neural network.

data. The higher-quality samples will be selected from these pseudo labels through a filtration process that considers the confidence levels of the soft labels produced by the teacher model. This process is known as self-training. This is a simple and effective method. However, there are still some problems. First, selecting pseudo labels may result in a great imbalance in data distribution, because the teacher model is relatively confident in the common samples. If the selection is based on the confidence of the generated soft labels, the distribution of the selected samples will be extremely imbalanced. Secondly, high confidence does not allow for sufficient training samples, and there is minimal utilization of unlabeled data. On the contrast, a low confidence reduces the quality of pseudo labels and causes a large quantity of noise. Therefore, the main difficulty in applying semi-supervised learning to automatic chord recognition is how to get the most from unlabeled data on the basis of introducing less noise.

In this paper, we firstly apply the contrastive learning, a discriminative representation learning framework to ACR. It is a frame work free from relying on high-quality pseudo labels and allows for a more comprehensive utilization of all unlabeled data. The algorithm proposed in this paper consists of two training phases. In the first phase of training, based on the contrastive learning, the model learns the general features of the dataset through discerning the connections and differences among different samples. We refer to the first phase of training as the representation learning phase. Contrastive learning makes comprehensive and equal use of unlabeled data. It does not focus on some unlabeled data, thus resulting in the underutilization of some high-quality unlabeled data with high confidence. Therefore, after the representation learning, these high-quality unlabeled data and labeled data will be used for semi-supervised learning for the model based on the noisy-student network. The algorithm for framework of the training strategy proposed in this article is shown in Fig. 3.

The algorithm proposed in this paper mainly consists of two stages: representation learning and semi-supervised learning. In representation learning, firstly, positive and negative sample pairs are constructed by data augmentation. The similar sample pairs are positive, and the dissimilar are negative sample pairs. Then, these sample pairs are input into the model to obtain the corresponding feature representations. Finally, the contrastive loss function is employed to pull together feature representations between positive sample pairs and push away that between negative sample pairs, so as to achieve the purpose of learning the general features of the dataset. In the process of semi-supervised learning which is based on the noisy- student framework, the model continues to be trained by using labeled data as well as high-quality unlabeled data. First, the model trained after the representation learning is used as the initial student model for this phase. Then the labeled data and the unlabeled data after augmentation are input to the student model to generate corresponding probability distribution of chords. The supervised loss and the unsupervised loss constitute the final loss, which is used to update the parameters of the student model. Representation learning facilitates the model to acquire a more comprehensively understanding from unlabeled data and explore deeper feature representations. The semi-supervised learning helps the model to make full use of high-quality unlabeled data. After two phases of learning, the model can fully utilize the unlabeled data and maximize the advantages of a large amount of unlabeled data. In the remaining part of Section III, a detailed description of these two phases is provided.

C. Representation Learning Based on Contrastive Learning

In this part, the representation learning phase based on contrastive learning is illustrated. The key to contrastive learning is how to construct positive and negative samples of a given sample. We first introduce the method to construct positive sample pairs, which is followed by the construction method for negative sample pairs. Finally, the method for constructing the contrastive loss function is introduced.

1) Method to Construct Positive Sample Pairs: In contrastive learning, similar samples are positive sample pairs. In image recognition tasks, the samples are usually augmented randomly to generate positive sample pairs. In other words, the original image is augmented through cropping or flipping as well as changing its brightness, which is used to construct a positive sample pair with the original image. This is followed by allowing the model to learn deeper features through pulling together the feature representations of the original sample and that of the sample after data augmentation. Therefore, in this paper, data augmentation is firstly employed to generate

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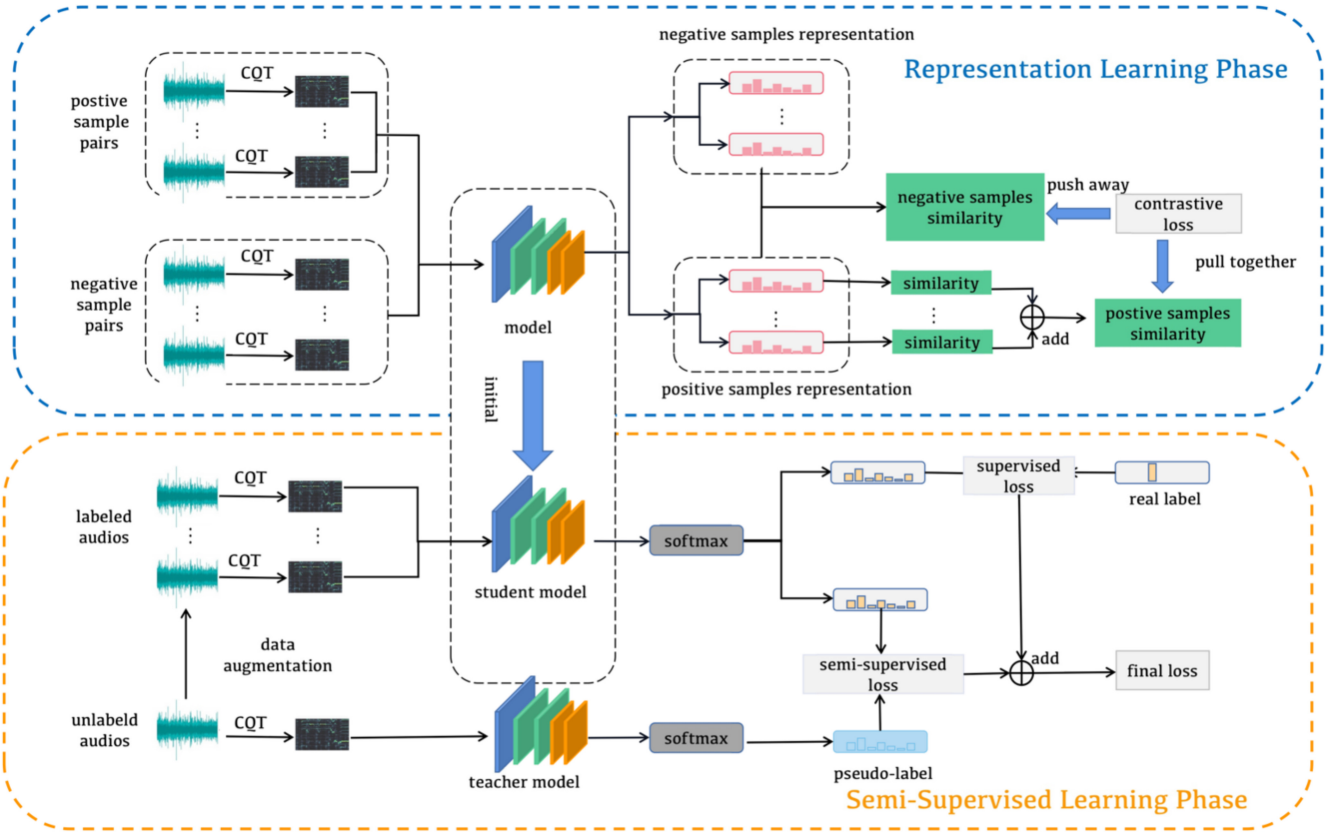


Fig. 3. Overview about the proposed algorithm.

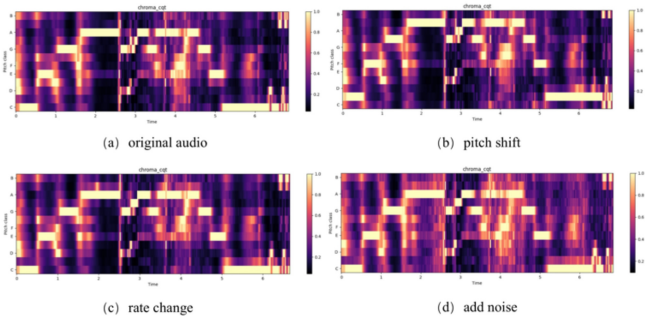


Fig. 4. Chromagram after data augmentation.

positive sample pairs. A clip of the audio is augmented by several common augmentation approaches.

The chromagram of the augmented audio is shown in Fig. 4. It can be seen that the chromagram of the audio with the shifted pitch (Fig. 4(b)) produces a relatively significant change compared to the chromagram of the initial audio. Pitch elements change vertically. The key to chord recognition is the analysis of pitch, and the shift in pitch prevents the chromagram from matching the real labels. The pitch-shifted audio and the original audio form two totally different chord categories. It does not meet the requirement of similarity between positive sample pairs, so augmenting the audio by shifting pitch is not feasible in constructing positive sample pairs. Changes to the audio rate (Fig. 4(c)) may cause the inconsistency of real labels between the augmented audio and the original one at a particular moment, which will be amplified as time goes by. Nevertheless, chord progressions are highly repetitive and a slight rate change would only cause a minimal label inconsistency. In addition, the audio used for training is not a complete piece of music but only a small part of the audio. Even if it is used to train the tail end of the audio, it would not have a great effect. Thus, the augmentation method such as a slight adjustment in audio rate is feasible in the construction of positive sample pairs. Fig. 4(d) shows that a small amount of Gaussian noise is added to the audio. It can be seen that the chromagram of augmented audio retains the key feature information, and adding noise can bring more obvious changes to the chromagram and produce more diversified positive sample pairs, which is beneficial to the learning of the model. Therefore, this augmentation method is also feasible.

In contrast to other tasks in music information retrieval, automatic chord recognition has a high requirement on data augmentation requirements due to its frame-by-frame nature.

Using more macroscopic augmentation methods is likely to damage the frame-level feature representation of the spectrogram, resulting in significant discrepancies between the augmented data and the real labels. Therefore, data augmentation methods that will cause minor variations are suitable for chord recognition. But it also brings a new problem: such positive sample pairs composed of the slightly augmented sample and the original sample cannot facilitate the model

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TABLE I
REAL CHORD SEQUENCE

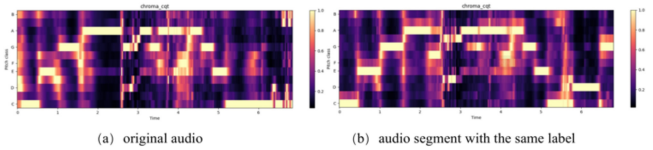
StartTime(s)	EndTime(s)	ChordType
20.662	22.346	C:maj
22.346	24.064	A:min
24.064	25.782	A:min
25.782	27.501	C:maj
27.501	29.219	C:maj
29.219	30.984	A:min
30.984	32.749	A:min
32.749	34.514	C:maj
34.514	36.279	A:min
36.279	37.973	F:maj
37.973	39.666	F:maj

to acquire enough information. Specifically, on one hand, the feature representation of the slightly augmented samples closely resembles that of the original sample. Pulling together feature representations of such two samples is not effective. On the other hand, the data augmentation method is not specifically designed for automatic chord recognition as what the model learns is the general representation of the audio without considering any chord-related information.

Therefore, the key to constructing positive sample pairs lies in maximizing the diversity of the positive pairs while ensuring the similarity among the positive samples. Obviously, relying only on the two data augmentation methods mentioned above to construct positive sample pairs will not allow the model to be trained well. In order to solve this problem, in addition to the data augmentation methods, we take different segments from the same audio to construct positive sample pairs. The real-label segment of the song "I Don't mind" is selected as shown in Table I.

Chord annotations are placed in the order of starting time, ending time, and chord category. Where the last chord ends is where the next starts. It can be seen that the chord progression exhibits a high level of repetition, which indicates that chord progressions are likely to be the same for different segments of the same song. It is mainly due to that a piece of audio often consists of several major chord categories that alternate throughout the audio in several minutes. Therefore, we consider truncating different segments from the same audio to construct positive sample pairs. Segments mentioned here are those with different starting times but same real labels in the audio. For example, for the real labels shown in Table I, the segment from 20.662(s) to 27.501(s) and the segment from 27.535(s) to 34.514(s) can be selected to form a positive sample pair. It is because both segments consist of chord labels C:maj and A:min in the completely same order. Each of them lasts almost for the same period of time and fully satisfies the requirement of using similar samples in constructing positive sample pairs. The chromagrams corresponding to the two audio segments are shown in Fig. 5.

It is evident that compared with other data augmentation methods, the positive sample pairs constructed by label-overlapping are relatively diversified while ensuring the similarity to the given sample. That is to say, compared with constructing positive sample pairs by augmenting the original audio, using samples with the same labels from different audio segments could maximize the diversity of positive sample pairs.



(a) original audio (b) audio segment with the same label

Fig. 5. Comparison of the chromagram between the original audio and audio with the same labels.

Such positive sample pairs can contain more useful information. However, for the unlabeled data, it is impossible to get real labels of audio, so we use pseudo labels instead of real labels to construct positive sample pairs in this paper. Specifically, for a given unlabeled sample $x_i = (x_i^1, x_i^2, \ldots, x_i^T)$ of length T, the segment x_i is selected from the song X. First, X is input to the pre-training model to generate pseudo labels of the whole song, and pseudo labels $y_i = (y_i^1, y_i^2, \ldots, y_i^T)$ corresponding to the segment x_i is intercepted. Then the pseudo-labeled segment $y_j = (y_j^1, y_j^2, \ldots, y_j^T)$ with the highest overlapping as the sequence y_i is selected from the rest of the pseudo labels of X. The corresponding audio segment x_j and the given sample x_i constitute a positive sample pair. The formula for calculating the overlapping between two samples is shown in (3):

$$overlap(x_i, x_j) = \frac{\sum_{t=1}^{T}\left(y_i^t \cap y_j^t\right)}{T} \tag{3}$$

However, there are two problems when using pseudo labels to construct positive sample pairs: ① For some given samples, it is difficult to identify a segment in the same song that exhibits a high degree of overlapping in its chord progression. That is to say, even the audio segment with the highest overlapping will be slightly different from the given sample. If the segment is directly used to form a positive sample pair with the given sample, it may affect what the model already learned. To address this issue, when quantifying the similarity of the positive sample pairs constructed by pseudo labels, it is considered to introduce overlapping of pseudo labels between two samples of the positive sample pair in this paper, i.e., the overlapping weight coefficient. When the overlapping of pseudo labels between two samples is higher than a given threshold, the overlapping weight coefficient is regarded as the overlapping between them when measuring the similarity of the positive sample pair. When the overlapping of pseudo labels between two samples falls below the given threshold, the overlapping weight coefficient will be 0, i.e., these two samples do not constitute a positive sample pair. The overlapping weight coefficient will deliver an important effect on the final loss function, and the specific formula will be presented in Section III-C3). ② Pseudo labels are not always correct, and incorrect pseudo labels may induce the model to misclassify samples from different categories as positive sample pairs, which will destroy what the model already

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learned. In this paper, the confidence factor is introduced to address this problem. Because positive sample pairs with higher confidence are more likely to be correctly categorized, thus we give greater weights to such positive sample pairs in our training. Specifically, assume that the predicted output of the pre-trained model about a frame x_i^t in the unlabeled sample x_i is $[p_t^1, p_t^2, \ldots, p_t^N]$, where N denotes the total number of chord categories and p_t^k denotes the probability of x_i^t being predicted as chord k. $p_t^{k_{max}}$ is assumed as the maximum value in $p_t^1, p_t^2, \ldots, p_t^N$. The model will eventually predict x_i^t as chord k_{max}, i.e., the pseudo label $y_i^t = k_{max}$. $p_t^{k_{max}}$ is the confidence of the model about the frame x_i^t. The larger it is, the more confident the model will be about the frame x_i^t, which means the prediction will more likely to be correct so that the pseudo label is more reliable. For the unlabeled sample x_i, its confidence factor of x_i is calculated in (4):

$$CFD(x_i) = \frac{p_{i,1}^{k_{max}} + p_{i,2}^{k_{max}} + \cdots + p_{i,T}^{k_{max}}}{T} \tag{4}$$

where $p_{i,t}^{k_{max}}$ denotes the confidence of frame t in the sample x_i. T denotes the total number of frames of the sample x_i. For the positive sample pair x_i and x_j constructed by pseudo-label method, its confidence weight in training is $CFD(x_i) \times CFD(x_j)$. The higher the confidence of the pre-trained model about samples x_i and $x_j$is, the more likely x_i and x_j will form correct positive sample pair. Its weight coefficient will be larger, and the impact on training will be more significant. Conversely, the weight coefficient is smaller, and it will not have a great impact on training even if it is an incorrect positive sample pair. The confidence weight of pseudo labels will significantly influence the final loss function, and the specific formulas are presented in Section III-C3). Except for using data augmentation, pseudo labels are applied to intercept the highly similar samples in the same audio to construct positive sample pairs. With these two approaches, positive sample pairs can be as diversified as possible while ensuring their similarity, thus allowing the model to learn more.

2) Method to Construct Negative Sample Pairs: In contrastive learning, dissimilar samples can constitute negative sample pairs. The simplest construction method is to form negative sample pairs constructed by each of the remaining samples in the same training batch with the current sample. However, some samples in the same batch are likely to be the same type of chord as the current sample, which results in these similar sample pairs being miscategorized as negative sample pairs. It is bad for the learning of model. In automatic chord recognition, due to the imbalanced distribution of different chords, there may be samples highly overlapped with the real label of the sample in the same training batch. If these samples are considered as negative samples of the current sample in the training, the model will push away feature representations of the same chords between both samples, potential undermining what the model already learned. In this paper, such samples such samples, which exhibit significant overlapping with the current sample but are misclassified as negative samples, are referred to as "false negative samples". Like the construction of positive sample pairs in Section III-C1), pseudo labels are employed to mitigate the impact of "false negative samples" on the model's performance in this paper. That is, pseudo labels are used to determine the overlapping between the remaining samples in the same training batch and the given sample. If the remaining sample is highly overlapped with the given sample, it indicates these two samples are similar and this sample is excluded from the calculation of the contrastive loss for the given sample. If there is no overlapping or the overlapping fall below the given threshold, the sample is suitable to be the negative sample of the given sample and included to the calculation of the contrastive loss of the given sample. Where the overlapping is calculated as in Section III-C1), and the specific calculation method is shown in (3).

3) Contrastive loss: Methods to construct positive and negative sample pairs were introduced in Section III-C1), and Section III-C2), respectively. This section is about how to construct a reasonable contrastive loss function that is used to narrow the gap of feature representations between positive sample pairs and widen the gap between negative sample pairs. The loss function in this paper is shown as follows:

$$Loss_{cl}(x_i) = -log\frac{sum\left(sim\left(x_i, x_i^+\right)\right)}{\sum_{j\neq i}^{2N-k} sim\left(x_i, x_j^-\right) + sum\left(sim\left(x_i, x_i^+\right)\right)} \tag{5}$$

where $Loss_{cl}(x_i)$ denotes the contrastive loss of the sample x_i, and $sum(sim(x_i, x_i^+))$ indicates the sum of the similarity of all positive sample pairs. $\sum_{j\neq i}^{2N-k} sim(x_i, x_j^-)$ denotes the sum of the similarity of all negative sample pairs. N denotes the total number of samples in the training batch of the sample x_i. k denotes the number of "false negative samples" excluded according to pseudo labels during constructing negative sample pairs, and its value is determined based on the specific training batch. It can be seen that this loss function is used to maximize the numerator and minimize the denominator with the purpose of maxmizing the similarity of positive sample pairs and minimize that of the negative sample pairs.

$$sum\left(sim\left(x_i, x_i^+\right)\right) = (1 - \mu * \alpha * \beta) * sim\left(x_i, x_i'\right) + \mu * \alpha * \beta * sim(x_i, x_s) \tag{6}$$

$$sim\left(x_i, x_j\right) = \exp\left(\frac{\cos(f(x_i), f(x_j))}{\tau}\right) \tag{7}$$

Eq. (6) illustrates how to calculate the similarity of positive sample pairs. Where x_i' denotes the sample obtained by performing two data augmentation on the sample x_i, and the specific data augmentation method is described in Section III-C1). x_s denotes the sample intercepted from the same song as the current sample which maintains the highest overlapping as the pseudo-label of the current sample. α denotes the weight factor used to determine whether the positive sample pairs constructed are similar or not as mentioned above, which serves to prevent the positive sample pairs being less overlapped with the pseudo labels and the selected positive sample pairs being less similar. β denotes the weight factor used to determine whether the positive sample pairs constructed are reliable or not as mentioned above, which serves to prevent the pseudo-label confidence from being too low and making the selection of positive sample pairs not

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reliable. μ denotes the balancing factor used to balance the two methods to construct positive sample pair. $f(x_i)$ denotes the feature representation obtained by the model, and the cosine similarity is used in this paper to calculate the similarity of two feature representation vectors. τ denotes the temperature coefficients. The specific formulas for calculating the pseudo-label overlap weight factor and the pseudo-label confidence weight factor are shown in (8) and (9), respectively:

$$\alpha = \begin{Bmatrix} overlap(x_i, x_s) & overlap(x_i, x_s) \geq \gamma \\ 0 & overlap(x_i, x_s) < \gamma \end{Bmatrix} \tag{8}$$

$$\beta = CFD(x_i) \times CFD(x_s) \tag{9}$$

where $overloap(x_i, x_j)$ denotes the pseudo-label overlap of samples x_i and x_j, and the calculation method is elaborated in (3). $CFD(x_i)$ denotes the confidence of the pseudo-label of the sample x_i, and the calculation method is detailed in (4).

D. Semi-Supervised Learning Based on Noisy-Student Network

The representation learning phase based on contrastive learning can fully utilize unlabeled data. However, some high-quality unlabeled data cannot be fully utilized. Therefore, we continue the semi-supervised training using labeled data and high-quality unlabeled data based on the model obtained in the first phase of training.

The first step is to select high-quality data from an extensive pool of unlabeled data. Specifically, a pre-trained teacher model $Model_T$ is firstly trained with labeled data, then the unlabeled sample x_i is input to $Model_T$ to obtain the corresponding sequence of chord probability, i.e., $[p^1_{i,1}, p^2_{i,1}, \ldots, p^N_{i,1}], \ldots, [p^1_{i,T}, p^2_{i,T}, \ldots, p^N_{i,T}]$. $p^k_{i,t}$ denotes the probability that the frame $\underline{\mathbf{t}}$ of the sample x_i is predicted to be chord k. Then the confidence of sample x_i is calculated according to the (4). If the confidence of sample x_i exceeds the predetermined threshold ω_0, sample x_i is classified as a high-quality sample. However, for samples containing rare chords, the confidence of these samples tends to be low. Setting a high threshold for all samples will result in an imbalanced distribution of filtered data. To solve this problem, the confidence threshold of the sample is determined based on (10):

$$\omega(x_i) = \omega_0 * \left(1 - \rho * \frac{t}{T}\right) \tag{10}$$

where $\omega(x_i)$ denotes the confidence threshold of sample x_i. t denotes the duration of the rare chord in sample x_i. T denotes the duration of sample x_i. ω_0 and ρ are hyperparameters. It can be seen that the longer the duration of a rare chord in a sample, the lower the confidence threshold for that sample will be. Compare the threshold $\omega(x_i)$ of the unlabeled sample x_i with its confidence $CFD(x_i)$. If $CFD(x_i)$ is greater than $\omega(x_i)$, x_i is classified as a high-quality sample. Otherwise, the sample will be excluded. The next step is to determine the strategy for semi-supervised learning. In this paper, we used the classical semi-supervised algorithm based on noisy-student network, which has achieved excellent performance in many areas. The specific training process is shown in Algorithm 1.

Algorithm 1 Semi-Supervised Training Steps

Input: labeled data X, true label Y, unlabeled data Z
Model: the pre-training teacher model T, the student model S
Functions: loss(CE_{loss}), confidence(CFD), data augmentation(Aug), back propagation (Bp)
Training process:

for $x \in X, y \in Y$
  do
  $p = T(x)$ — get the prediction of the teacher model on sample x
  $l = CE_{Loss}(p, y)$ — get the loss between the teacher model's prediction and real label
  $T \leftarrow Bp(T, l)$ — update the parameters of the teacher model
  end do
end for
for $x \in X, y \in Y, z \in Z$
  do
  $p_1 = S(x)$ — get the prediction of the student model on labeled sample x
  $l_1 = CE_{Loss}(p_1, y)$ — get the loss between the student model's prediction and real label
  $\varphi = T(z)$ — get the pseudo-label of the teacher model for unlabeled sample z
  if $CFD(\varphi) > \sigma$ — whether the confidence of pseudo label is higher than threshold
  $z' = Aug(z)$ — perform data augmentation on unlabeled sample z
  $p_2 = S(z')$ — get the prediction of student model on the augmented sample z'
  $l_2 = CE_{Loss}(p_2, \varphi)$ — get the loss between student model's prediction and pseudo-label
  $S \leftarrow Bp(S, l_2)$ — update the parameters of the student model
  else
  $S \leftarrow Bp(S, l_1)$ — update parameters of the student model with supervised loss only
  end do
end for

IV. Experiments

A. Dataset and Process

In order to ensure the fairness and validity of experimental results, the training data employed in this paper are all from uspopdataset2002[1] [19], a public dataset in automatic chord recognition. 2,076 popular songs were collected in this paper, including 185 annotated songs and 1,891 unlabeled songs due to content limitation and copyright issues. The label files contain information about the starting and ending time of the song, as well as the labeled chord progressions. However, due to copyright issues, it was not possible to obtain the audio completely aligned with the tag file, i.e., when the chord of the audio starts was slightly different from that of the tag file. To minimize the experimental error, the method proposed in [18]

[1]The annotations for uspop2002 are provided by the Music and Audio Research Lab at NYU.

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is adopted in this paper to manually adjust the time when the chord starts, only by adding or deleting the tag 'N' at the beginning of the tag file. We divided the labeled training data into the training dataset and validation dataset in an 80%:20% ratio. For the test dataset, we used the Beatles dataset to yield a cross-dataset experiment. We adopted the chord label mapping rules used in [9], and all types of chords were mapped to the vocabulary V including following 14 chord qualities: maj, min, 7, maj7, min7, minmaj7, sus4, sus2, min6, maj6, dim, aug, dim7 and hdim7. 169 types of chords could be recognized by the model. Complex chord qualities such as 9, maj9, maj11, min9, min11 and transposed chords are mapped into the lexicon V according to the corresponding rules.

In the input data processing stage, all audio files used for training are transformed into spectrograms through CQT by librosa [20] in this paper, which spans 6 octaves with a sampling rate of 22,050HZ and a hopsize of 2,048. Then the CQT features are converted into log expressions, which are calculated in (11):

$$S_{log} = In(S + \varepsilon) \tag{11}$$

where S denotes the CQT features, and ε is an extra small positive number to prevent outputting zero in logarithmic computation. Then, normalize through the whole process to reduce the gap between spectral energy of different audio segments. The specific method to calculate is shown in (12):

$$S_{norm} = \frac{S_{log} - mean(S_{log})}{var(S_{log})} \tag{12}$$

where $mean(S_{log})$ denotes the average value and $var(S_{log})$ denotes the variance. Before training, we shift the pitch of both labeled and unlabeled data from -5 semitones to $+6$ semitones and shift that of the CQT spectrogram directly as the augmented feature. For the labeled data, their corresponding chord labels are also shifted accordingly.

Both contrastive experiments and ablation experiments of ACR based on contrastive learning and noisy-student are based on the convolutional neural network augmented by bidirectional self-attention proposed in this paper. The source code is made public at Gitee.[2] Model training is optimized by using Adam optimizer with the initial learning rate at 0.0001. The training is divided into two phases, and the first phase is the representation learning phase based on contrastive learning. In this phase, only unlabeled-data are used for pre-training, with the output of the final hidden layer serving as the feature representation of the model. To construct positive sample pairs through data augmentation more randomly, we perform random data augmentation on all unlabeled data before each epoch. The range of random augmentation is represented by hyperparameters a and b in Table II, which determine the range of changing audio rate and noise augmentation, respectively. After the first training phase, the student model is initialized with parameters of the model obtained and the second training phase based on noisy-student starts. The values of all hyperparameters in this paper can be found in Table II.

[2]https://gitee.com/jjjjjjy/CL-NS-ACR/

TABLE II
EXPERIMENTAL PARAMETERS

Part	Parameters	Range
Bidirectional attention-augmented convolutional neural network	feature size	144
	input length	108
	self-attention heads	4
	dimension of Q, K, V	144
	attention layer repetition	4
Representation learning	rate change	0.95~1.05
	add noise	0.004~0.008
	temperature(τ)	0.2
	overlapping threshold(γ)	0.5
	balancing factor	0.5
Noisy-student network	ω_0	0.75
	ρ	0.5

B. Evaluation Metrics

For a better comparison with existing methods, we used the individual evaluation metrics from mir_eval [21], the most commonly used one in the automatic chord recognition task. These metrics are calculated based on the average frame-level accuracy:

$$acc_{frame} = \frac{\sum_{i=1}^{N} C_i}{\sum_{i=1}^{N} F_i} \tag{13}$$

where N denotes the number of songs used to evaluate the performance of the model. F_i denotes the duration of the i_{th} song. C_i is the duration of the model's correct prediction about the i_{th} song. The specific evaluation metrics in mir_eval include: Root Scores, Maj-min Scores, Seventh Scores, Thirds Scores, Triad Scores, Tetrads Scores, Mirex Scores, and all above seven metrics are used for evaluation in this paper.

C. Comparative Experiments

We first compare the model proposed in this paper with the classical neural networks in ACR. CNN is a deep convolutional neural network proposed in [22] for automatic chord recognition, and CNN-CRF is a conditional random field connected to the output of the convolutional neural network to smooth the sequence output of CNN. CRNN is a convolutional recurrent neural network used in [23], which shares the same idea as the model proposed in this paper. The convolutional layer is used to extract local features of the spectrogram, while the recurrent neural network is used for sequence feature extraction. CRNN-CRF is the conditional random field that is connected after the output of CRNN. BTC is an automatic chord recognition model based on bidirectional self-attention proposed in [18]. The bidirectional self-attention layer used in this paper is the same as the BTC structure.

The experimental results are shown in Table III. It can be seen that the proposed model achieves competitive results on all evaluation metrics. CNN_AUG significantly outperforms BTC that is only based on self-attention. This suggests that the capability to capture short-term features of convolutional

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TABLE III
COMPARISON OF DIFFERENT MODELS

Model	Root	Thirds	Triads	Seventh	Tetrads	Maj-min	MIREX
CNN[22]	78.63	75.25	73.02	64.23	60.26	76.62	76.06
CNN+CRF[22]	79.17	75.71	73.42	64.77	60.83	**77.34**	76.31
CRNN[23]	78.70	74.93	72.85	63.95	59.96	76.44	76.01
CRNN+CRF[23]	77.98	74.66	72.67	64.39	60.43	76.21	75.49
BTC[18]	78.71	75.45	73.31	64.94	61.00	76.85	76.03
BTC+CRF[18]	78.65	75.18	72.91	64.94	61.02	76.38	75.59
CNN_AUG(ours)	78.82	**75.80**	73.59	65.30	61.32	77.20	76.24
CNN_AUG+CRF	**79.33**	75.75	**73.60**	**65.58**	**61.66**	77.05	**76.39**

TABLE IV
COMPARISON OF DIFFERENT METHODS

Methods	Root	Thirds	Triads	Seventh	Tetrads	Maj-min	MIREX
baseline	78.82	75.80	73.59	65.30	61.32	77.20	76.24
S-T	79.51	76.15	73.95	65.81	61.94	77.37	76.99
SSL	79.23	76.03	73.75	65.63	61.67	77.34	76.77
CL+NS	**79.80**	76.28	74.13	**66.17**	62.13	77.70	77.00
CL+NS+ST	79.59	**76.51**	**74.21**	66.12	**62.15**	**77.73**	**77.32**

TABLE V
RESULT OF ABLATION EXPERIMENTS

Methods	Root	Thirds	Triads	Seventh	Tetrads	Maj-min	MIREX
baseline	78.82	75.80	73.59	65.30	61.32	77.20	76.24
NS	79.18	76.08	73.66	65.47	61.48	77.30	76.58
CL	79.34	76.25	74.05	65.68	61.69	77.59	76.84
CL_{aug}	79.41	76.12	73.91	65.50	61.55	77.46	76.60
CL+NS	**79.80**	**76.28**	**74.13**	**66.17**	**62.13**	**77.70**	**77.00**

neural networks is crucial for the data-poor task of ACR. It is worth noting that the model proposed in this paper achieves the best performance after the applications of CRF.

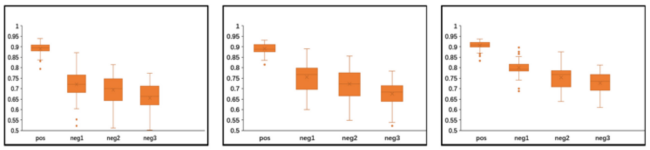
We compare the semi-supervised automatic chord recognition algorithm based on the contrastive learning and noisy-student network proposed in this paper with several methods that have performed well in automatic chord recognition. We chose a structured algorithm for decomposing chord labels proposed in [23], which decomposes chord labels according to components such as root tone and chord quality, and then recodes the chord labels. It performs well in ACR. In addition, we choose the semi-supervised chord recognition algorithm proposed by [10] as the comparative algorithm in this paper. It is worth noting that we modify this algorithm so that it no longer focuses on rare chords, which allows a better comparison with our algorithm. The experimental results are shown in Table IV.

Where baseline denotes the model trained only under supervision. S-T denotes the model trained based on the structured training algorithm proposed in [23]. SSL denotes the model trained based on the semi-supervised learning algorithm proposed in [10]. CL+NS denotes the model trained based on the contrastive learning and noisy-student network proposed in this paper. It can be seen that CL+NS outperforms baseline and SSL in all evaluation metrics, and it is on par with S-T. This indicates that the algorithm proposed in this paper significantly improves the performance of the model. The algorithm proposed in this paper can be well combined with S-T and achieves the best performance, which also shows the good adaptability of the proposed algorithm.

D. Ablation Experiments

We conduct detailed ablation experiments on the proposed algorithm, focusing on the construction method of positive and negative sample pairs in representation learning as well as the role of each stage of training. The experimental results are shown in Table V. Where baseline denotes the model trained only under supervision. NS denotes the model trained based on the noisy- student network only. CL denotes the model trained based on the contrastive learning only. CL_{aug} denotes the construction of positive sample pairs by data augmentation only. It can be seen that both CL and NS have improved compared with baseline and the combination of these two methods delivers the most excellent results. We believe this is thanks to the method proposed in this paper, and the contrastive learning and the noisy-student network are complementary. Contrastive learning allows for more comprehensive use of unlabeled data and thus learning general features of the dataset. Noisy-Student network, on the other hand, makes full use of high-quality unlabeled data. Two methods have different emphases on unlabeled data and

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(a) τ =0.1 (b) τ =0.2 (c) τ =0.8

Fig. 6. Comparison of similarity between positive and negative samples with different temperature coefficients.

therefore complement each other well to achieve the best performance. It can be seen that CL_{aug} is improved compared with baseline, but still lower than CL.

This indicates that the strategy proposed in this paper to construct positive and negative samples based on pseudo-labeling is effective, and this method can help the model perform better by maximizing its diversity while ensuring the similarity of positive sample pairs.

E. Parametric Analysis

We analyze the temperature coefficients used in the contrastive loss function, i.e., parameters τ in (7). After the experiment, we find that the model can perform optimally when $\tau = 0.2$. To further validate the association between different temperature coefficients and the similarity of positive and negative samples, we draw a graph that depicts the similarity between positive and negative samples in the same training batch at different temperature coefficients, as illustrated in Fig. 6. Pos in the horizontal axis of Fig. 6. indicates the positive samples of the current sample. Neg1, neg2, and neg3 indicate three negative samples with the highest similarity to the current sample. It can be seen that when $\tau = 0.1$, there is a large gap in the similarity of the feature representations between the positive sample pair and negative sample pair, which indicates that This suggests that the model gives larger gradients to these negative samples that are similar to the current sample. This also confirms that the contrastive loss function is a difficult sample self-discovery loss function. When the temperature coefficient is small, the model pays extra attention to the negative samples that are more similar to the current sample.

However, the experimental results are not satisfactory when $\tau = 0.1$. This is mainly because there are obvious dependencies and correlations among structures of different chords, for example, maj7 and maj with the same root note should be more similar in feature representation. A small temperature coefficient will focus more on the difference between these similar negative samples without paying attention to the correlation among them, and this extreme case will also hinder the learning of the model. When $\tau = 0.8$, the gap in the similarity between positive samples and negative sample pairs is small, which indicates that the role of the contrastive loss function in difficult sample self-discovery is substantially weakened at this time, and the model treats all negative samples equally during the training process and no longer pays extra attention to those negative samples that are highly similar to the current sample. When the gap in the distribution of similarity between positive and negative sample pair is at a moderate position, the contrastive loss function can avoid more extreme cases while taking into account the strong self-discovery effect of difficult samples.

V. Conclusion

In this paper, we first present a self-attention-augmented convolutional neural network that achieves competitive performance on automatic chord recognition. Then we analyze the strengths and weaknesses of current semi-supervised algorithms and the difficulties in applying semi-supervised algorithms to automatic chord recognition. We also propose a semi-supervised automatic chord recognition algorithm based on contrastive learning and noisy-student network, which fully and comprehensively utilizes a large amount of unlabeled data to improve the performance of the model. Our proposed algorithm is comparable to the state-of-the-art automatic chord recognition algorithms and can be adapted to other algorithms. We will further explore the integration with more algorithms in future work.

But our method also has limitations. The three-stage model proposed in this paper has redundancy in structure, and further analysis is needed to extract feature information from the hidden layers of each part and fuse and filter the network layers. Although the algorithm proposed in this paper improves the generalization of the model, it is still limited by the impact of data distribution differences. Moreover, the scale of unlabeled data used in this paper is relatively small, and future work can attempt to conduct experiments based on more unlabeled data.

In future work, we will further investigate the impact of other data enhancement methods on experimental results. On the other hand, we will also try to apply more classical semi-supervised algorithms to automatic chord recognition, and we believe that the application of semi-supervised algorithms will bring more surprises for this task.

References

[1] H.-G. Kim, G. Y. Kim, and J. Y. Kim, "Music recommendation system using human activity recognition from accelerometer data," *IEEE Trans. Consum. Electron.*, vol. 65, no. 3, pp. 349–358, Aug. 2019, doi: 10.1109/TCE.2019.2924177.

[2] J. A. Burgoyne, J. Wild, and I. Fujinaga, "An expert ground truth set for audio chord recognition and music analysis," in *Proc. 12nd ISMIR*, Miami, FL, USA, 2011, pp. 633–638.

[3] J. Pauwels, K. O'Hanlon, E. Gómez, and M. B. Sandler, "20 years of automatic chord recognition from audio," in *Proc. 20th ISMIR*, Delft, The Netherlands, 2019, pp. 54–63.

[4] J. E. van Engelen and H. H. Hoos, "A survey on semi-supervised learning," *Mach. Learn.*, vol. 109, pp. 373–440, Nov. 2020, doi: 10.1007/s10994-019-05855-6.

[5] T. Fujishima, "Realtime chord recognition of musical sound: A system using common lisp music," in *Proc. 25th ICMC*, Beijing, China, 1999, pp. 464–467.

[6] H.-G. Kim, G. Y. Lee, and M.-S. Kim, "Dual-function integrated emotion-based music classification system using features from physiological signals," *IEEE Trans. Consum. Electron.*, vol. 67, no. 4, pp. 341–349, Oct. 2021, doi: 10.1109/TCE.2021.3120445.

[7] S.-C. Lim, J.-S. Lee, S.-J. Jang, S.-P. Lee, and M. Y. Kim, "Music-genre classification system based on spectro-temporal features and feature selection," *IEEE Trans. Consum. Electron.*, vol. 58, no. 4, pp. 1262–1268, Nov. 2012, doi: 1109/TCE.2012.6414994.

[8] J. Deng and Y. K. Kwok, "Large vocabulary automatic chord estimation with an even chance training scheme," in *Proc. 18th ISMIR*, Suzhou, China, 2017, pp. 531–536.

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[9] L. Rowe and G. Tzanetakis, "Curriculum learning for imbalanced classification in large vocabulary automatic chord recognition," in *Proc. 22nd ISMIR*, 2021, pp. 586–593.
[10] M. Bortolozzo, R. Schramm, and C. R. Jung, "Improving the classification of rare chords with unlabeled data," in *Proc. IEEE ICASSP*, Toronto, ON, Canada, 2021, pp. 3390–94.
[11] Y. Wu, T. Carsault, E. Nakamura, and K. Yoshii, "Semi-supervised neural chord estimation based on a variational autoencoder with latent chord labels and features," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 2956–66, Oct. 2020, doi: 10.1109/TASLP.2020.3035001.
[12] Y. Wu and W. Li, "Automatic audio chord recognition with MIDI-trained deep feature and BLSTM-CRF sequence decoding model," *IEEE Trans. Audio, Speech Lang. Process.*, vol. 27, no. 2, pp. 355–366, Feb. 2019, doi: 10.1109/TASLP.2018.2879399.
[13] Y. Wu and K. Yoshii, "Joint chord and key estimation based on a hierarchical variational autoencoder with multi-task learning," *APSIPA Trans. Signal Inf. Process.*, vol. 11, p. e19, Mar. 2022, doi: 10.1561/116.00000052.
[14] A. Jaiswal, A. R. Babu, M. Z. Zadeh, D. Banerjee, and F. Makedon, "A survey on contrastive self-supervised learning," *Technologies*, vol. 9, no. 1, p. 2, Dec. 2020, doi: 10.3390/technologies9010002.
[15] S. Kum, J.-H. Lin, L. Su, and J. Nam, "Semi-supervised learning using teacher-student models for vocal melody extraction," in *Proc. 21st ISMIR*, 2020, pp. 531–536.
[16] Q. Xie, M.-T. Luong, E. Hovy, and Q. V. Le, "Self-training with noisy student improves ImageNet classification," in *Proc. IEEE CVPR*, Seattle, WA, USA, 2020, pp. 10684–95.
[17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, Las Vegas, NV, USA, 2016, pp. 770–778.
[18] J. Park, K. Choi, S. Jeon, D. Kim, and J. Park, "A bi-directional transformer for musical chord recognition," in *Proc. 20th ISMIR*, Delft, The Netherlands, 2019, pp. 620–627.
[19] A. Berenzweig, B. Logan, D. Ellis, and B. Whitman, "A large-scale evaluation of acoustic and subjective music-similarity measures," *Comput. Music J.*, vol. 28, no. 2, pp. 63–76, 2004, doi: 10.1162/014892604323112257.
[20] B. McFee, "librosa/librosa: 0.7. 1," Oct. 2019.
[21] C. Raffel et al., "MIR_EVAL: A transparent implementation of common MIR metrics," in *Proc. 15th ISMIR*, 2014, pp. 367–372.
[22] F. Korzeniowski and G. Widmer, "A fully convolutional deep auditory model for musical chord recognition," in *Proc. IEEE 26th MLSP*, 2016, pp. 13–16.
[23] J. Jiang, K. Chen, W. Li, and G. Xia, "Large-vocabulary chord transcription via chord structure decomposition," in *Proc. 20th ISMIR*, Delft, The Netherlands, 2019, pp. 644–651.



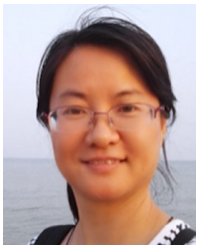
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